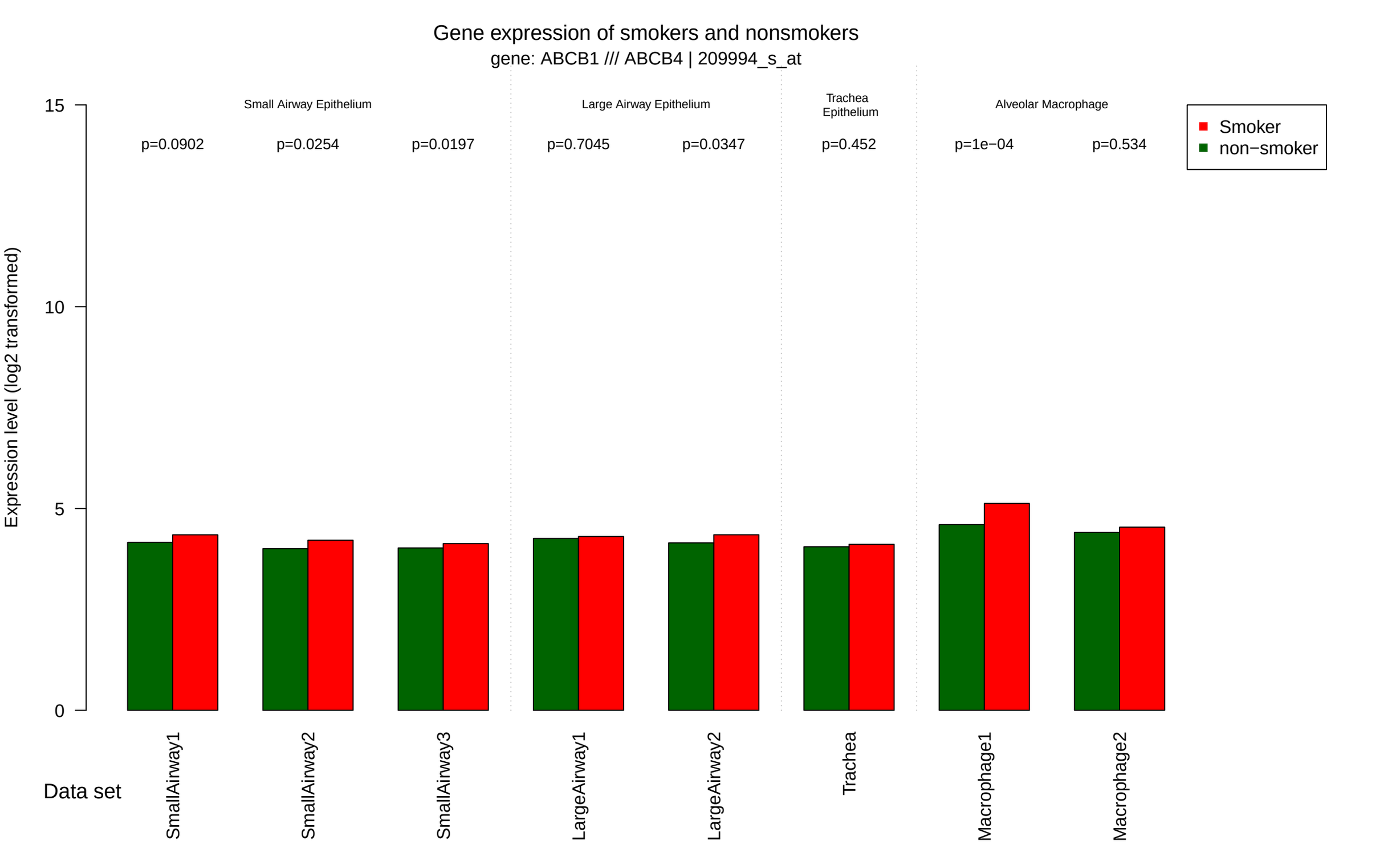
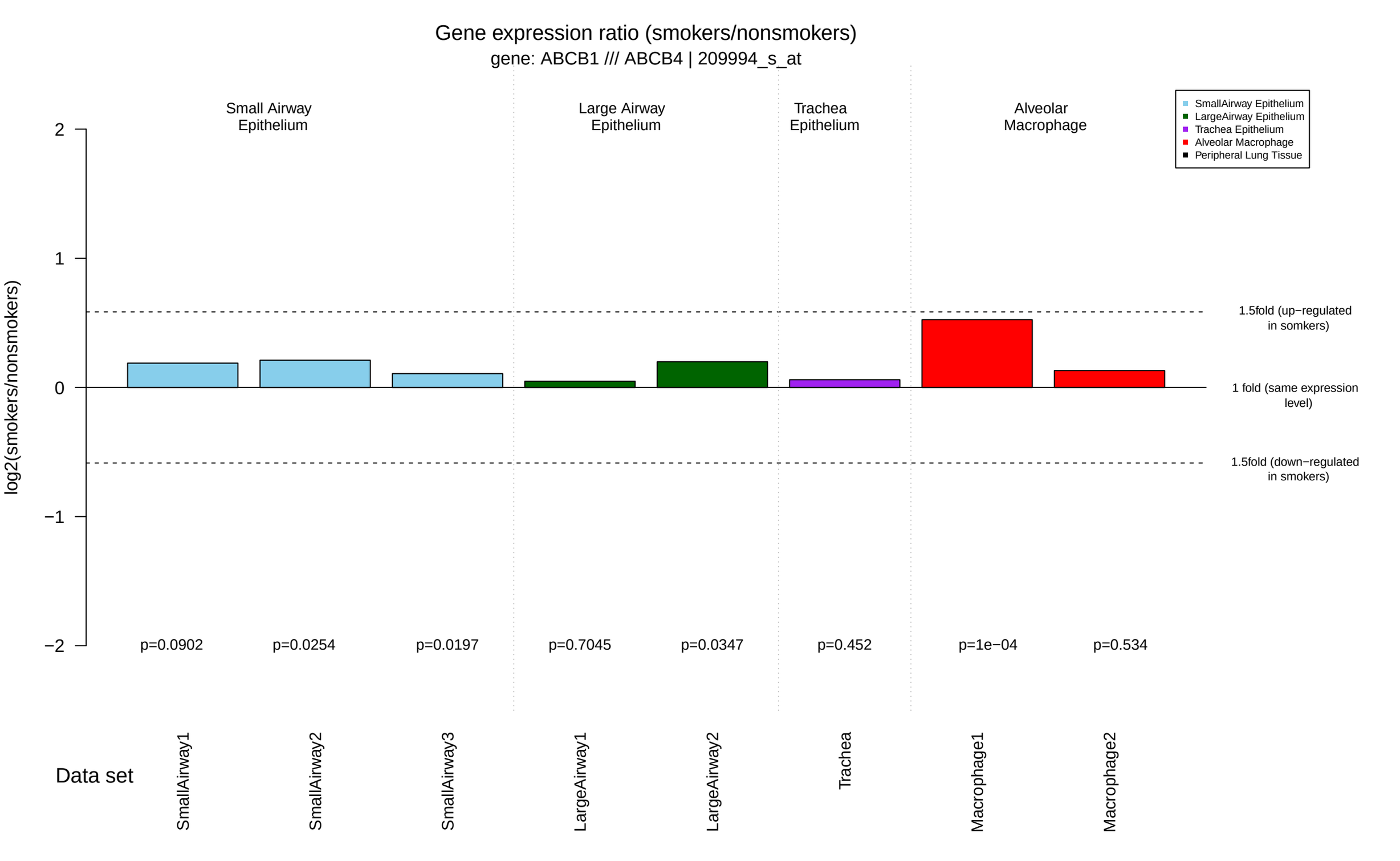
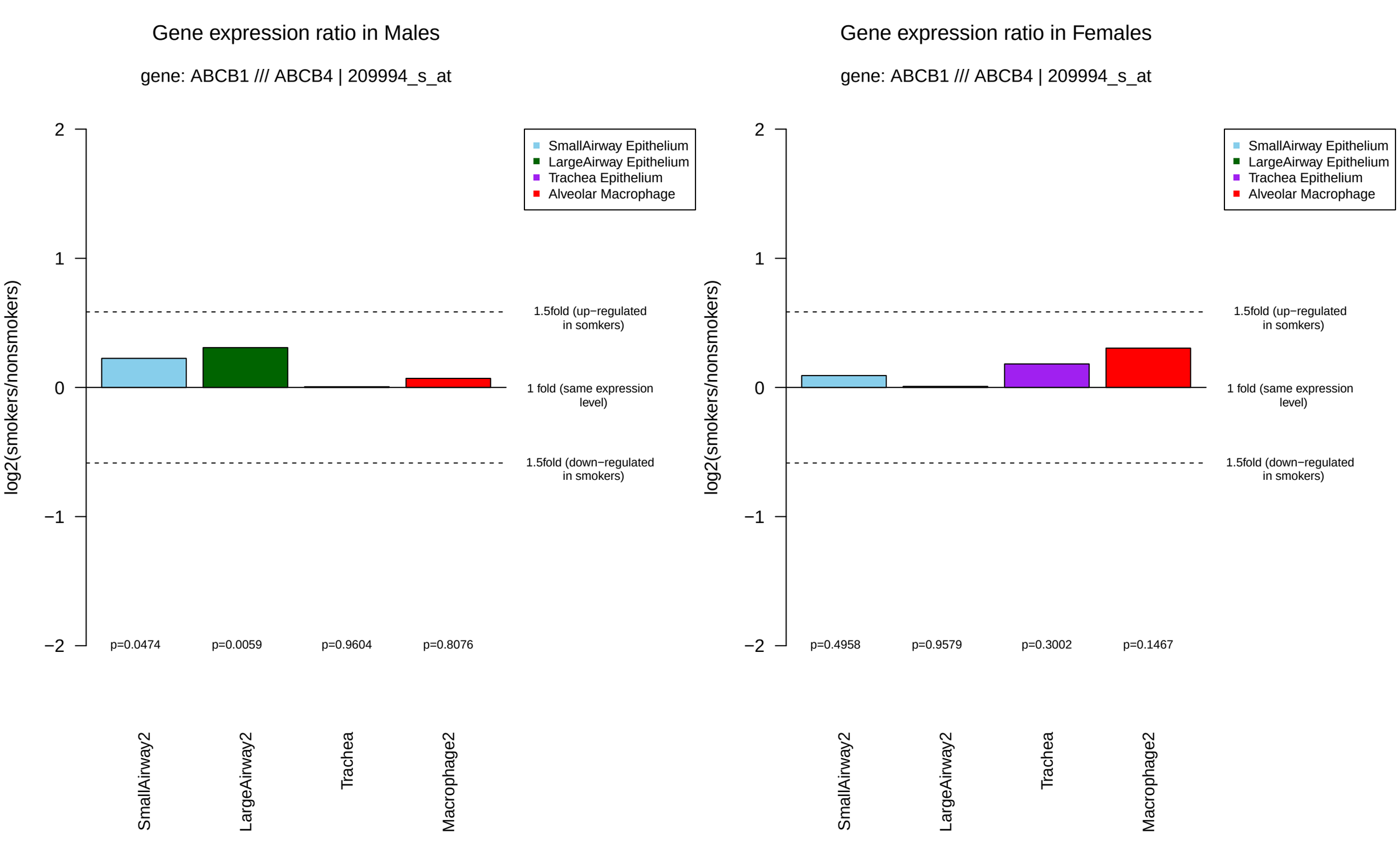
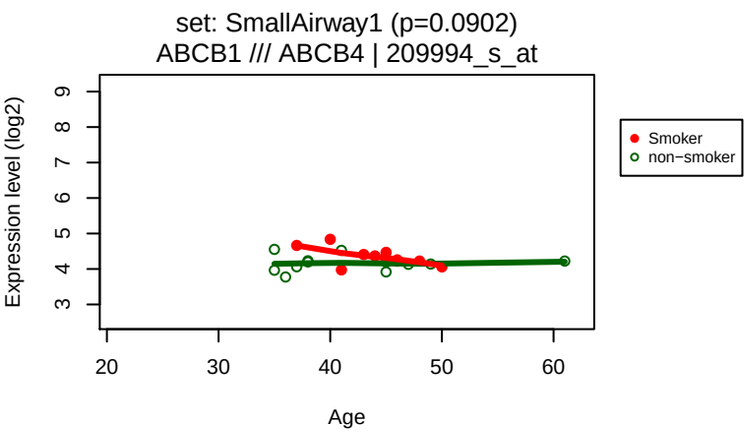
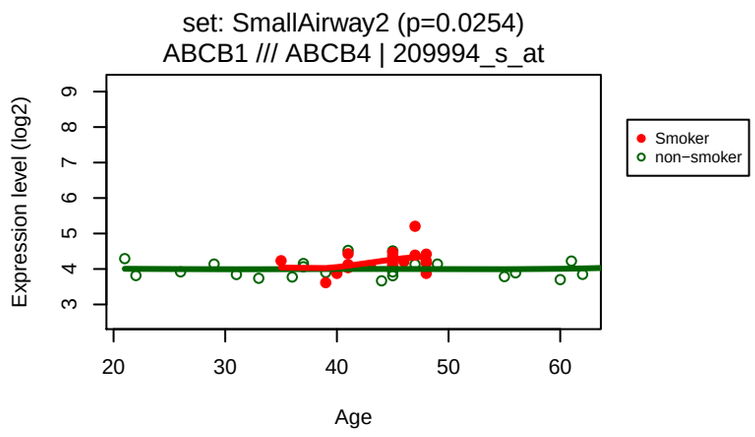
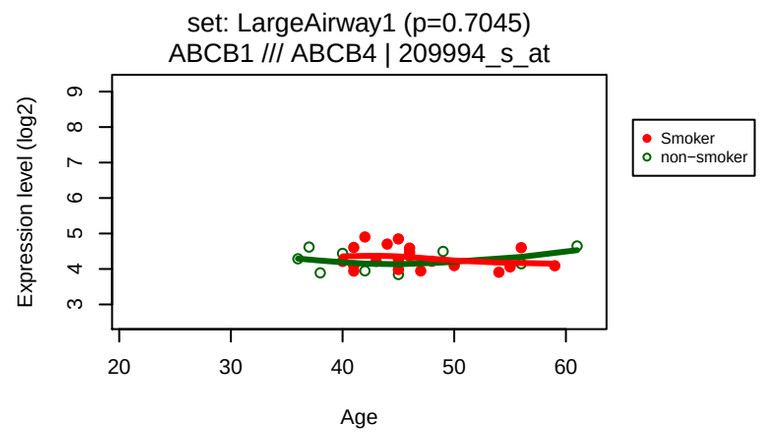

Gene expression of smokers and nonsmokers
gene: ABCB1 /// ABCB4 | 209994_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0902
p=0.0254
p=0.0197
p=0.7045
p=0.0347
p=0.452
p=1e−04
p=0.534
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

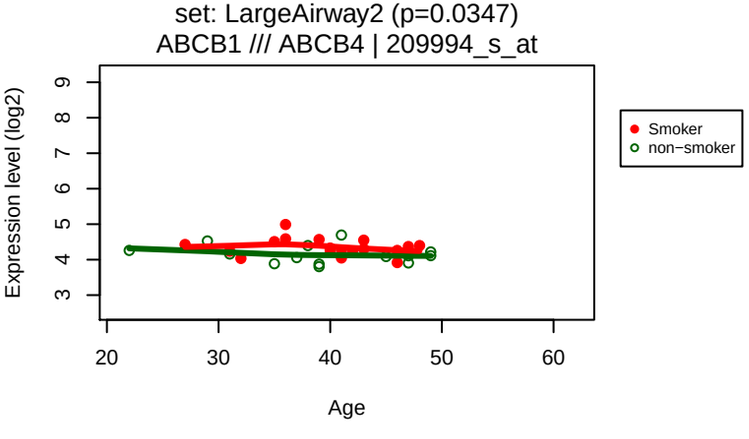

Gene expression ratio (smokers/nonsmokers)
gene: ABCB1 /// ABCB4 | 209994_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0902
p=0.0254
p=0.0197
p=0.7045
p=0.0347
p=0.452
p=1e−04
p=0.534
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

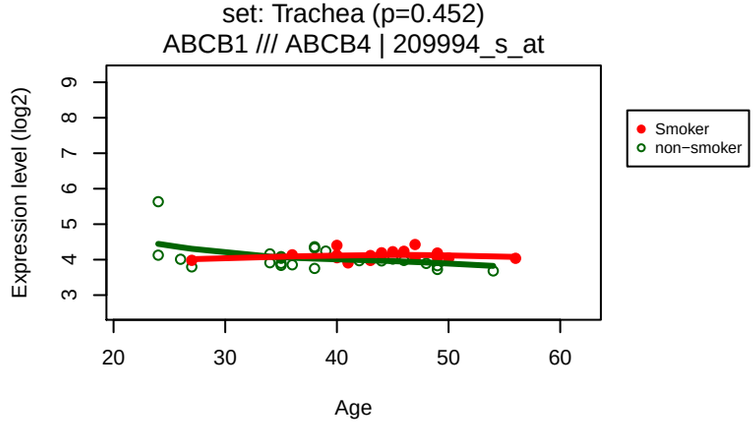

Gene expression ratio in Males
gene: ABCB1 /// ABCB4 | 209994_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0474
p=0.0059
p=0.9604
p=0.8076
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ABCB1 /// ABCB4 | 209994_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4958
p=0.9579
p=0.3002
p=0.1467
SmallAirway2
LargeAirway2
Trachea
Macrophage2

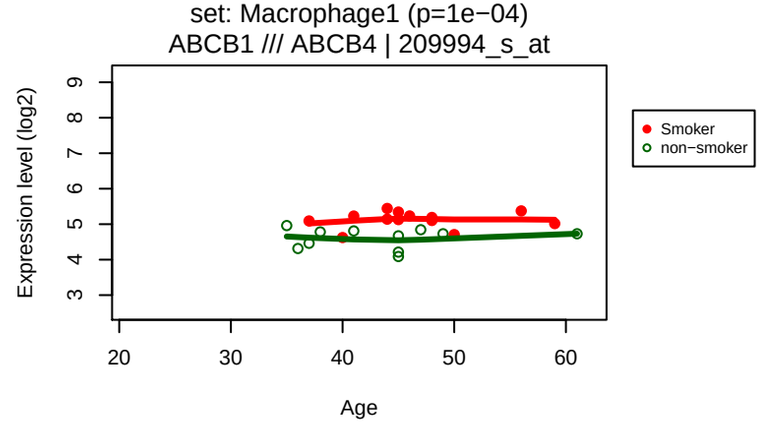

set: SmallAirway1 (p=0.0902)
ABCB1 /// ABCB4 | 209994_s_at
Expression level (log2)
Age
Smoker
non−smoker

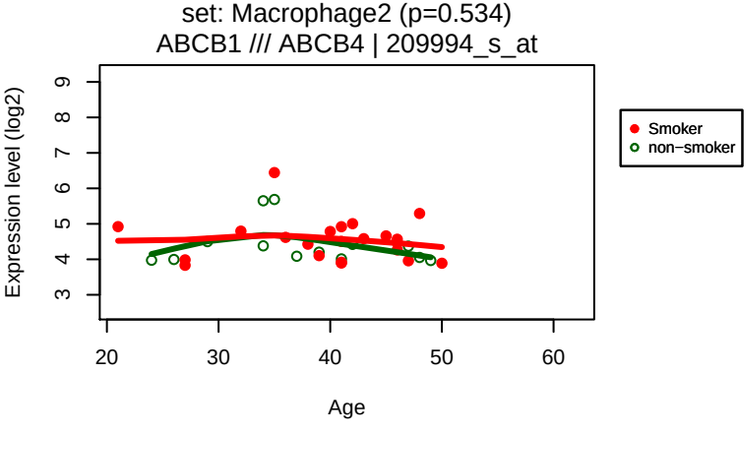

set: SmallAirway2 (p=0.0254)
ABCB1 /// ABCB4 | 209994_s_at
Expression level (log2)
Age
Smoker
non−smoker

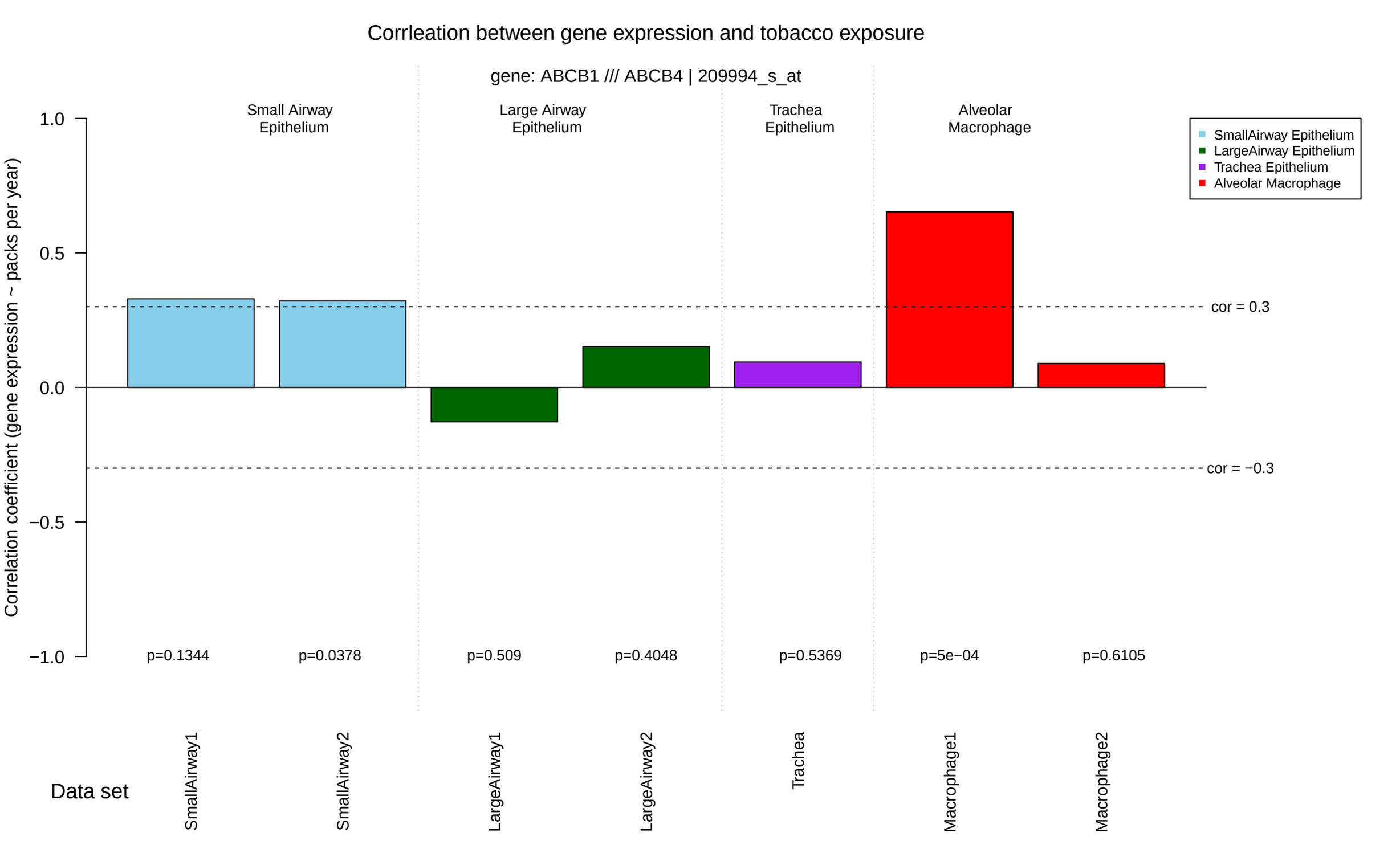

set: LargeAirway1 (p=0.7045)
ABCB1 /// ABCB4 | 209994_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0347)
ABCB1 /// ABCB4 | 209994_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.452)
ABCB1 /// ABCB4 | 209994_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=1e−04)
ABCB1 /// ABCB4 | 209994_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.534)
ABCB1 /// ABCB4 | 209994_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ABCB1 /// ABCB4 | 209994_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1344
p=0.0378
p=0.509
p=0.4048
p=0.5369
p=5e−04
p=0.6105
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2